IN104:

Monte Carlo Simulation of the Ising Model

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Git commands

- **git clone <url>**

  - Copy a remote repository to your local machine

- **git status**

  - Check current changes and branch state

- **git add <filename>**

  - Add local file for the next commit

Git commands

- **git commit -m "description"**
  - save changes with a description
- **git push**
  - upload commits to the remote repository
- **git pull**
  - collect changes from the remote repository

Monte Carlo Simulations

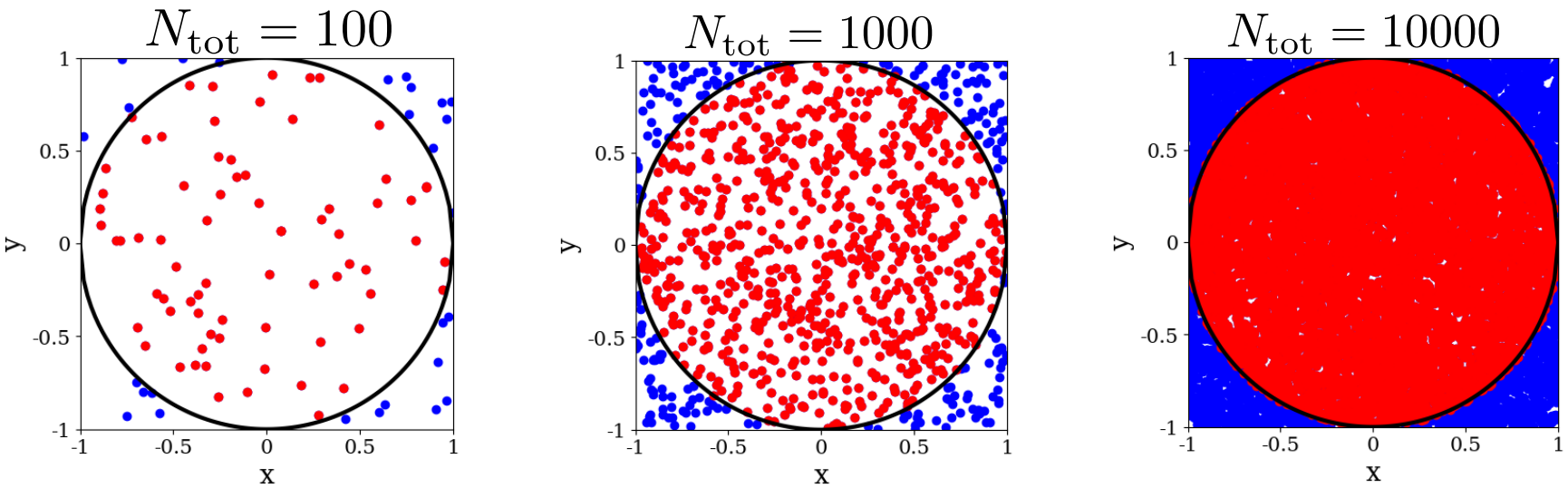
- Group of stochastic methods to simulate complex systems
  - the interaction of many components produce emergent behaviors
  - e.g. physical systems, financial markets, traffic flow, weather systems, ...

- Idea: use Random Numbers
  - to explore different configurations and
  - estimate properties that can not be easily obtained analytically.

Random Numbers: examples

- Sample random points within a square to estimate π

$$\frac{N_{\text{red}}}{N_{\text{tot}}} \approx \frac{\pi}{4}$$

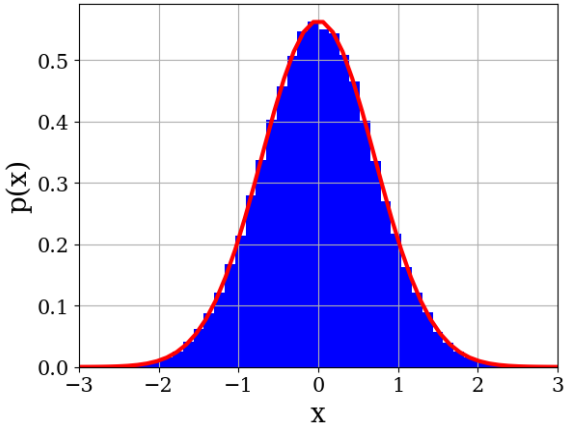


Random Numbers: examples

- Calculate complicated integrals $$I = \int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}$$

- Sampling probability distributions:

$$p(x) \propto e^{-x^2}$$



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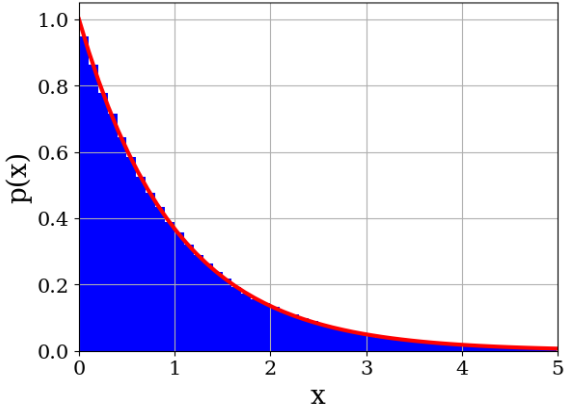


Accept / Reject

- Method to sample a probability distribution p(x)
  - Uniformly sample X in the domain
  - Uniformly sample Y in the range (0, max(p(x)))
  - If $y < p(X)$ accept X
  - Else reject;

Statistical Physics at Equilibrium

- Physical systems want to **minimize the energy** (cost function);
- Temperature causes stochastic noise (disorder);
- At equilibrium, physical systems follow the **Boltzmann probability distribution**

$$P(\{s\}) = \frac{1}{Z} e^{-E(\{s\})/T}, \qquad Z = \sum_{\{s\}} e^{-E(\{s\})/T}$$

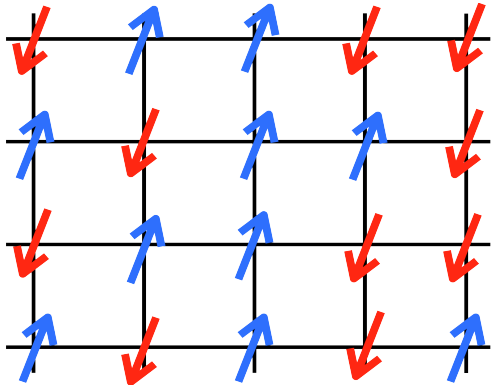
Q: what are the limits at **zero** and **infinite** temperature?

The Ising Model

- A minimal model to explain a magnetic phase transition
  - from paramagnetic (high T) to ferromagnetic (low T)

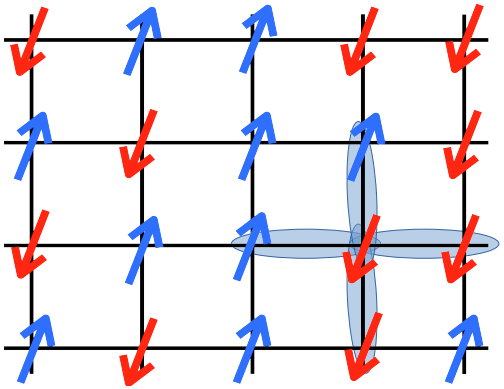
- The macroscopic magnetization is determined by the collective behavior of local magnetic moments (spins)

- Spins $\{S_i\}$ can be either up (+1) or down (-1)



The Ising Model (J>0)

- Energy: $$E(\{S_i\}) = -J \sum_{\langle i,j \rangle} S_i S_j$$



- Where $\langle i, j \rangle$ means that spins only interact with nearest neighbors.

- Q: How many possible configurations for a L x L square lattice ?

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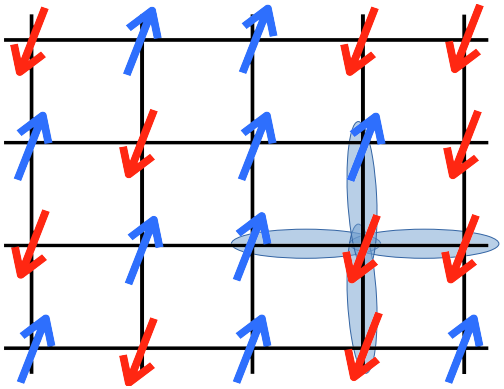

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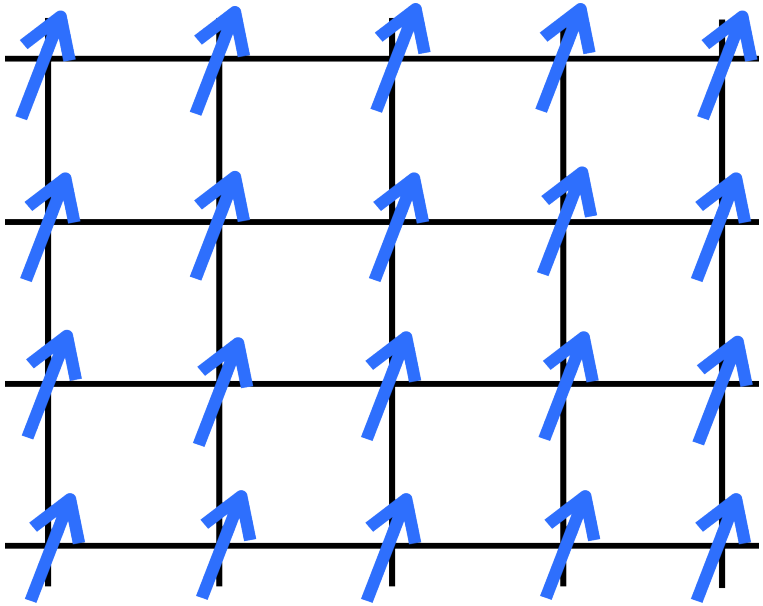
- Q: How many possible configurations for a L x L square lattice ?

$2^{L^2} \rightsquigarrow$ Evaluating Z becomes impossible even for small L

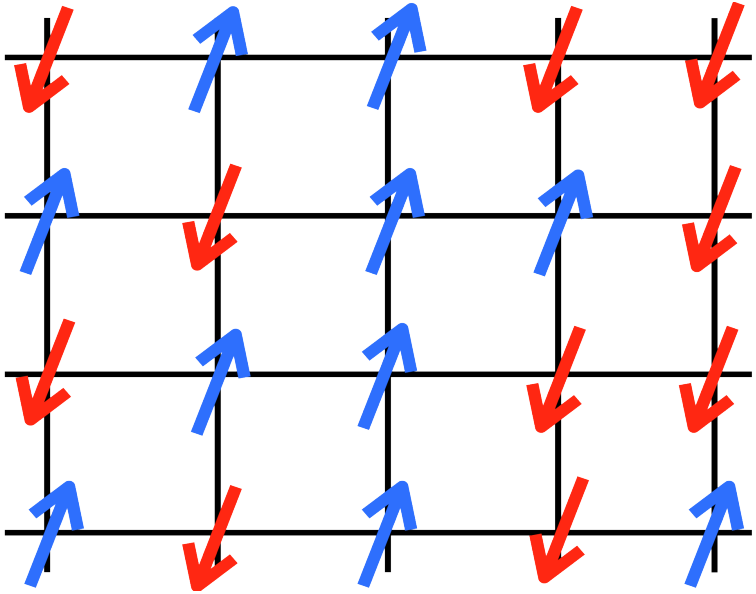
Phase transition

Average magnetizaion $m = \frac{1}{L^2} \sum_i S_i$

$T < T_c,\ m \approx 1(\text{or} - 1!)$



$T > T_c,\ m \approx 0$



Metropolis algorithm

- Efficient way to sample the Boltzmann probability distribution

  - (1) Start from an initial configuration $\{S_i^{(0)}\}$
  - (2) Compute its energy E_0
  - (3) Select a spin at random and propose to flip it
  - (4) Compute the new energy E_n
  - (5) Accept the move:
    - always if $E_n < E_0$
    - otherwise with probability $e^{-(E_n - E_0)/T}$
  - (6) Update the configuration accordingly
  - (7) Measure observables and repeat the process (go back to step (3))

Metropolis algorithm

- It can be shown that the sampled states will follow the **Boltzmann probability distribution:**

  - **Ergodicity** (all possible configurations can be reached) ***
  - **Detailed balance** ***

- Once the state is generated, we can calculate observables (energy, magnetization, **squared magnetization**, ...):

Postprocessing

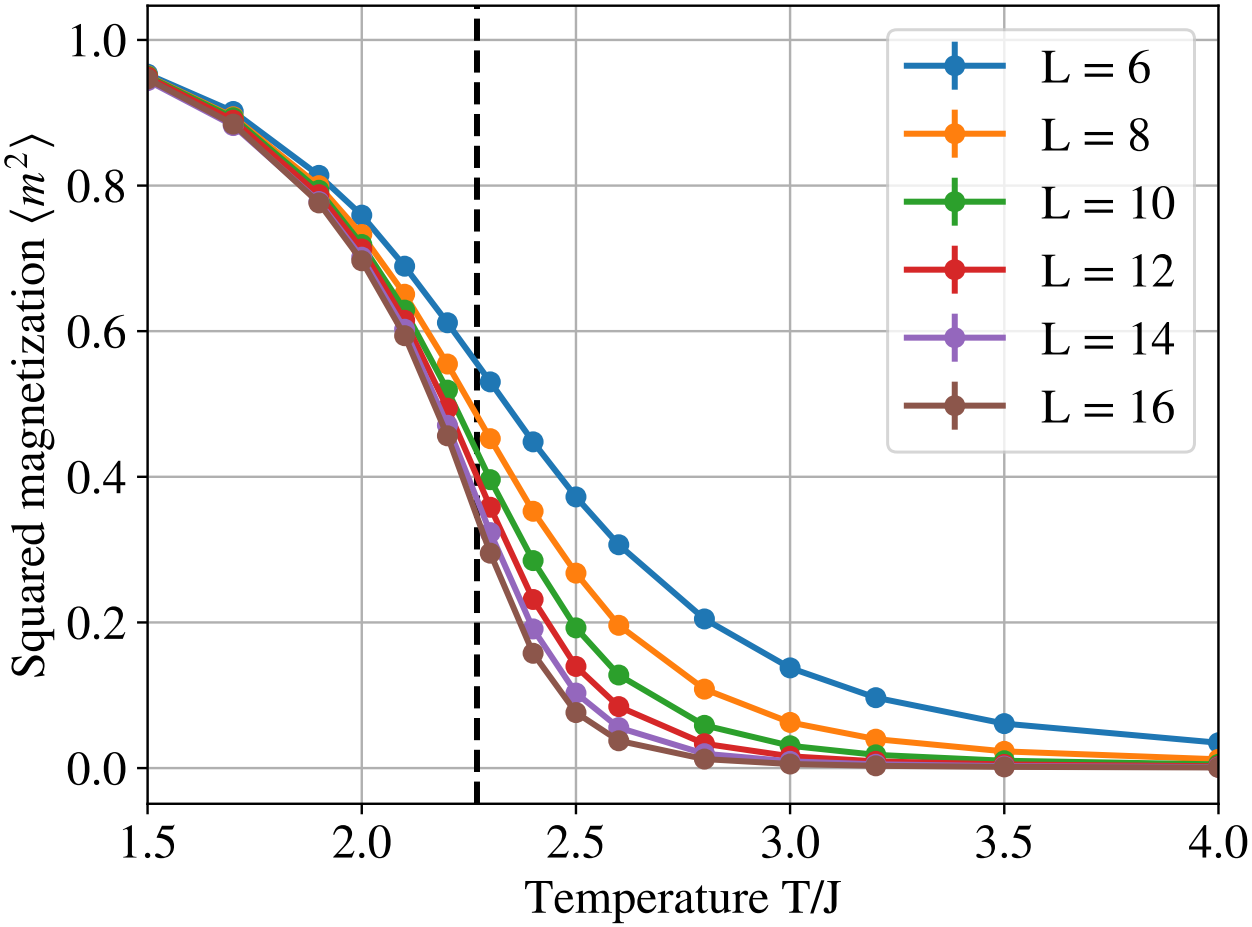
- We generated N samples following Boltzmann p.d.f;

- We extract the value of observable O calculating the mean on all samples:

$$\langle O \rangle \approx \frac{1}{N} \sum_{n=0}^{N} O_n$$

- And the standard deviation $\sigma_O \approx \sqrt{\frac{1}{N-1} \sum_{n=0}^{N} (O_n - \langle O \rangle)^2}$

Results: squared magnetization

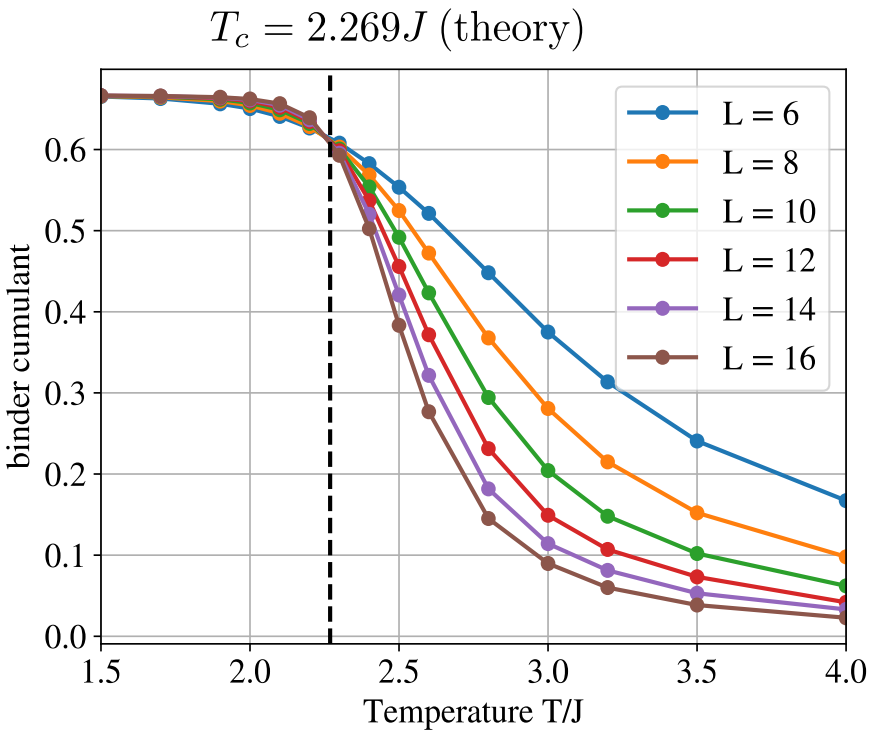


$T_c = 2.269J$ (theory)

Results: Binder cumulant

Scale invariant quantity at the critical point Tc

$$b = 1 - \frac{\langle m^4 \rangle}{3 \langle m^2 \rangle^2}$$



Next time

- Autocorrelations
- Finite-size scaling
- Swendsen – Wang Algorithm

Weakness of Metropolis: Autocorrelations

- The current configuration is different from the previous one by at most one single spin

- Subsequent configurations are strongly correlated

- Risk of not exploring all possible states $\rightsquigarrow$ the desired p.d.f is not well sampled

- This issue is important near the phase transition: critical slowing down

Autocorrelation function

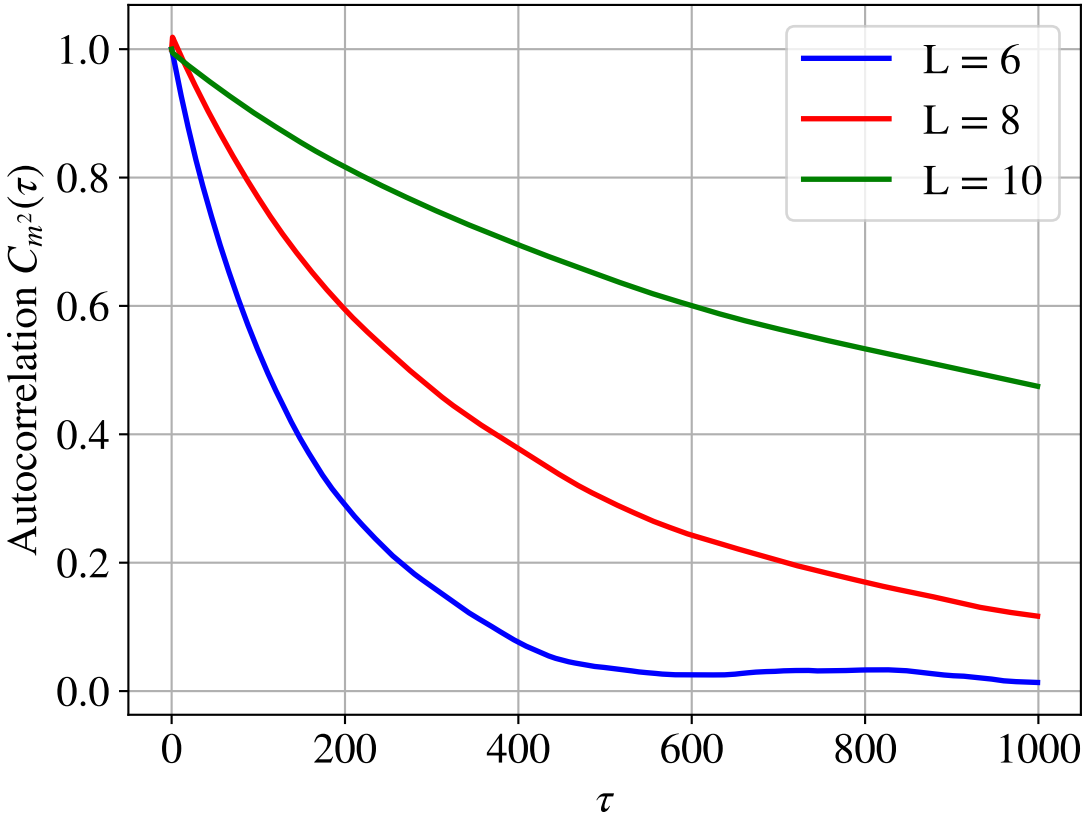
- Given the observable O (e.g. magnetization)

$$C(\tau) = \frac{\langle O(t+\tau)O(t)\rangle - \langle O(t+\tau)\rangle \langle O(t)\rangle}{\sigma_O^2}$$

- Where t is the step in the chain.

- If $|C(\tau)| \approx 1$, strong autocorrelation ; if $|C(\tau)| \approx 0$ weak autocorrelation

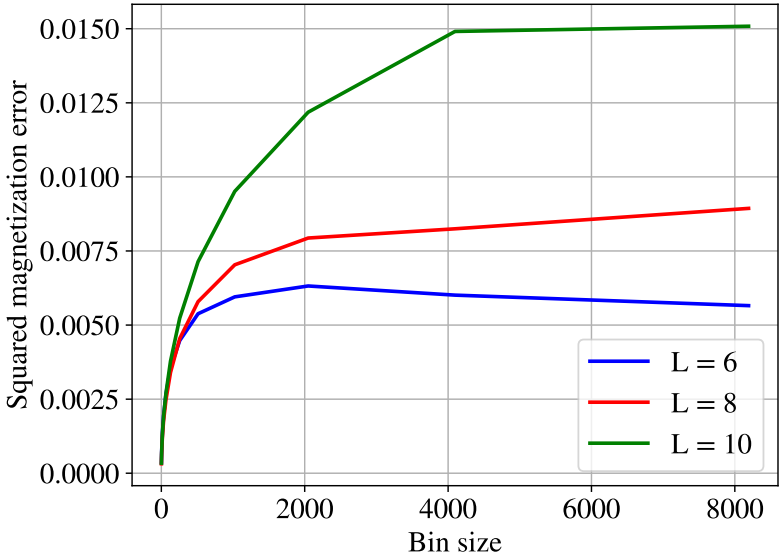
Autocorrelation function



Autocorrelation

- If autocorrelation is large, errors are underestimated

- Solution: binning analysis
  - Group data points in N bins of size d;
  - Error = standard deviation of mean values of all bins
  - Check various bin sizes until convergence



Low T and T

- At low T, it is very unlikely that a single spin flips;
- The system is frozen in one of the two configurations (m=+1 or m=−1)
- It means that Boltzmann distribution is not correctly sampled

Swendsen-Wang Algorithm

- Flip clusters of spins at once → reduce autocorrelations
  → correctly sample the Boltzmann distribution
  → Less iterations are needed

Evaluation

Minimum requirements:

core algorithm: basic working Metropolis algorithm;

pre- and post- processing: statistical tools (mean, std, ...) and write/read to/from file.

- Bonus points

  - autocorrelations and binning analysis; **

  - binder cumulants, suscettivity, ... **

  - code clarity, indentation, comments, ...

- Top:

  - Efficiency;

  - Swendsen-Wang algorithm **

** explained next time